

Empirical Scale Invariance of Multiscale Coherence Across 31 Orders of Magnitude

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Abstract

When does a measurable quantity begin to resemble a state variable rather than a domain-specific observable? We evaluate this question for multiscale coherence, quantified by the fixed functional $C[\Psi] = (r_{\text{micro}} + r_{\text{macro}} + r_{\text{meso}})/3$, where the three components summarize local correlation strength, global synchrony, and hierarchical network compactness. Across 18 real-data domains spanning approximately 31 orders of magnitude in characteristic timescale, the discriminative performance of the framework shows no detectable association with timescale under pre-specified correlation tests (Pearson $\rho = -0.002$, $p = 0.994$; Spearman $\rho = -0.018$, $p = 0.945$; Kendall $\tau = -0.014$, $p = 0.939$; bootstrap 95% CI $[-0.45, +0.40]$). A null-model comparison shows that this scale invariance is not shared by simple control observables, which remain near chance (mean AUC ≈ 0.56) while the multiscale framework yields mean AUC ≈ 0.85 . A temporal coarse-graining analysis on 13,257 twelve-lead ECGs reveals a monotonic increase in classification performance from static windows to sub-second resolution (AUC $0.697 \rightarrow 0.795$; Spearman $\rho = +0.943$, $p = 0.005$), consistent with a multiscale organizational signal rather than a single-scale feature. Finally, the product $C[\Psi] \cdot T[\Psi]$ shows descriptive quasi-invariance within several regime families, although the evidence remains limited and is presented as suggestive rather than formal. These results support a narrower claim than fundamentality: multiscale coherence behaves as a cross-domain, empirically robust organizational coordinate that warrants further theoretical and experimental study. We specify explicit falsification criteria and release all code and datasets for independent replication.

1. Introduction

Physical state variables are rarely recognized all at once. They typically begin as observables that prove unusually stable across contexts before receiving a deeper formal interpretation. Temperature evolved from a subjective sensation to a thermoscopic reading to Boltzmann's $\partial S/\partial E$ over two centuries. Entropy was an engineering quantity before it became a cornerstone of statistical mechanics. Information was a communication concept before Landauer and Bekenstein connected it to thermodynamics.

The present paper asks a narrower question: does multiscale coherence behave empirically more like a domain-specific metric, or more like a transferable organizational coordinate? The functional $C[\Psi]$, introduced in [P1] and validated across seven empirical domains, computes the mean of three correlation-based observables: r_{micro} (mean absolute pairwise correlation), r_{macro} (RMS correlation), and r_{meso} (Mantegna MST compactness). Its companion $T[\Psi]$ measures inter-scale tension; $S[\Psi]$ quantifies internal fragility.

We do not claim that $C[\Psi]$ is a fundamental dimension. We test whether it satisfies empirical criteria one would expect from a candidate state variable: fixed measurability, non-triviality relative to simpler descriptors, cross-domain transferability, and scale invariance. Throughout this paper, the main empirical object is the full multiscale framework (C , T , S), not $C[\Psi]$ alone.

2. Evaluation Criteria

(i) Fixed measurability. Identical protocol across all domains: same window, normalization, MST algorithm. No domain-specific tuning.

(ii) Non-triviality. The framework should not collapse to any single tested component or simple summary statistic. In [P1], single components achieve lower performance than the composite, and random-matrix nulls remain separated. This establishes non-triviality relative to tested alternatives, not irreducibility in the formal physical sense.

(iii) Cross-domain transferability. Discriminating structured from unstructured states across unrelated systems without re-training.

(iv) Scale invariance. No systematic dependence of discriminative value on characteristic timescale.

3. Framework Summary

$r_{\text{micro}} = (2/N(N-1)) \sum_{\{i < j\}} |\rho_{\{ij\}}|$; $r_{\text{macro}} = \sqrt{(\text{mean}(p^2))}$; $r_{\text{meso}} = 1 - \bar{L}_{\text{MST}}/2$. $C[\Psi] = (r_{\text{micro}} + r_{\text{macro}} + r_{\text{meso}})/3$. $T[\Psi] = |r_{\text{micro}} - r_{\text{meso}}| + |r_{\text{meso}} - r_{\text{macro}}| + |r_{\text{micro}} - r_{\text{macro}}|$. $S[\Psi] = (\sigma_{\text{sub}}(r_{\text{micro}}) + \sigma_{\text{sub}}(r_{\text{meso}}))/2$. Full definitions in [P1].

4. Datasets: 18 Real-Data Domains

Table 1 lists the 18 real-data domains. An additional 20 simulation-calibrated domains serve as controls but do not enter the core claim. All datasets are publicly available.

Domain	Field	n	Value	Stat	Scale	Tier
ECG AF (MIT-BIH)	Cardiology	29	0.961	AUC	~ms	Pillar
ECG static (Chapman)	Cardiology	13,257	0.713	AUC	~10 ms	Pillar
Sleep EEG	Neuroscience	22,233	449.2	F	~0.1 s	Pillar
Finance S&P 500	Economics	4,750	0.718	AUC	~day	Pillar
Music genres	Acoustics	250+	43.6	F	~s	Pillar
Genomics BRCA	Biology	286	95.3	t	~Myr	Pillar
Seismic precursors	Geophysics	11	p=.057	p	~yr	Pillar
Gait Parkinson	Medicine	165	0.819	AUC	~s	Strong
Exoplanets Kepler	Astrophysics	1,433	0.899	AUC	~day	Strong
Galaxies SDSS	Cosmology	62	0.901	AUC	~Gyr	Strong
Heat Waves EU	Climate	5,297	0.714	AUC	~day	Strong
Art perception	Psychophysics	20	0.933	AUC	~s	Strong
Solar System	Astronomy	8	0.795	AUC	~kyr	Extension
Gaia clusters	Astrophysics	66	0.765	AUC	~Myr	Extension
Sunspot 400yr	Solar physics	1,812	0.777	AUC	~yr	Extension
Ocean ENSO	Oceanography	208	0.603	AUC	~yr	Extension
Climate modes	Meteorology	841	0.827	AUC	~yr	Extension
Energy grid EU	Engineering	2,190	0.767	mean	~day	Extension

Table 1: Complete real-data inventory. Tier: Pillar ($n > 200$, $AUC > 0.70$ or $F > 40$), Strong (significant, smaller n), Extension (smaller effect, included for scale coverage). Timescales are order-of-magnitude assignments.

5. Test 1: Scale Invariance

Across the 18 real-data domains, the estimated association between discriminative performance and characteristic timescale is near zero under all pre-specified tests, with no indication that any single domain drives the result:

Pearson $\rho = -0.002$ ($p = 0.994$). Spearman $\rho = -0.018$ ($p = 0.945$). Kendall $\tau = -0.014$ ($p = 0.939$). Bootstrap 95% CI: $[-0.45, +0.40]$. Leave-one-out: removing any domain yields $|\rho| < 0.15$.

The timescale range spans approximately 31 orders of magnitude from millisecond cardiac intervals to gigayear galactic structure timescales. Modest reassignments of order-of-magnitude timescales would not materially alter these results.

6. Test 2: Null Model Comparison

Scale invariance could reflect a property of the evaluation protocol rather than of the framework. We compute five control observables (mean variance, spectral entropy, first eigenvalue ratio, kurtosis, lag-1 autocorrelation) on identical data with identical windowing. Controls achieve mean $AUC \approx 0.56$ (range: 0.48–0.65); $C[\Psi]$ achieves mean $AUC \approx 0.85$ (range: 0.60–0.96). The gap is consistent and robust to subsampling. Conditional on the ablation results in [P1], the multiscale construction is necessary for the observed performance.

7. Test 3: Temporal Coarse-Graining

On 13,257 Chapman-Shaoxing ECGs at six coarse-graining levels: AUC increases monotonically from $L0 = 0.697$ to $L5 = 0.795$ (Spearman $\rho = +0.943$, $p = 0.005$). The enrichment from static to sub-second windows is clear, followed by a plateau at the finest scale. This is consistent with a multiscale signal distributed across nested temporal resolutions. Individual components do not show this monotonicity.

8. Descriptive Quasi-Invariance of $C \times T$

We report the product $C[\Psi] \cdot T[\Psi]$ as a descriptive regularity, not as a conservation law. Within dynamical regimes: Plein Sync ratio = 0.895 ($p = 0.492$), Fragmentation 1.004 ($p = 0.963$), Critical 1.072 ($p = 0.222$). The null hypothesis of invariance cannot be rejected within regimes. However, $C \times T$ is not conserved across domains (inter-domain CV = 34.2%), and per-regime sample sizes remain small. This is best interpreted as a hypothesis-generating constraint on (C, T) geometry.

9. Discussion

9.1 What Would Strengthen the Case?

A companion paper [P3] now provides a candidate evolution equation $dC/dt = -\lambda(\Omega)[C - C_{eq}]$ with damped Rabi oscillations ($R^2 = 0.46$), and identifies a variational principle (action minimality at 88–100%). What would further strengthen the case: formal derivation of the

potential from first principles, rigorous characterization of (C, T, S) geometry, independent replication on pillar domains, and demonstration of a domain where the framework fails informatively.

9.2 Limitations

(1) No first-principles derivation. (2) Timescales are order-of-magnitude assignments. (3) Simulation-calibrated domains do not carry the core claim. (4) C×T evidence is suggestive only. (5) Single research group; independent replication essential. (6) Requires temporal depth (PTB-XL static AUC = 0.46).

9.3 Falsification

Falsified if: (a) $|p| > 0.30$ on 25+ real domains; (b) a single-scale observable matches performance; (c) independent replication fails on any pillar domain; (d) coarse-graining gradient reverses on a second large-n dataset.

10. Conclusion

Multiscale coherence behaves as a transferable organizational coordinate whose discriminative value is robust across scales, domains, and temporal resolutions. It is not trivially reducible to simpler descriptors, and its scale invariance is not a generic property of summary statistics. Whether this earns the status of a genuine state variable will depend on derivation, replication, and experimental control. We release all code and datasets to facilitate this process.

References

- [P1] Fogel-Reinert. Universal Coherence Functional. Phys. Rev. E (2026). DOI:10.5281/zenodo.19444148.
- [P3] Fogel-Reinert. Multiscale Coherence Under Decoherence Reveals a Continuous Classical-Quantum Order Parameter. In preparation for PRL.
- [P7] Fogel-Reinert. Initial-State Multiscale Coherence Predicts Quantum Thermodynamic Efficiency. In preparation for PRL.
- [Code] Repository: <https://github.com/XXXXXX/cpsi-multiscale-coherence> — DOI: 10.5281/zenodo.XXXXXX (to be completed before submission).